Location: Toronto, Canada
Status: Married, with children

\I had knee surgery a few weeks ago because my knee pain was unbearable. As suggested by my doctor, I had knee replacement surgery. Now, I am following some prescribed exercises but I'm worried that I am doing them wrong and possibly slowing down recovery. On top of that, I have a vacation with my family and want to be the 'fun grandpa'".

- Goals:
  - To gain mobility and strength in his knee for an upcoming vacation and keep up with his active grandchildren.
  - He wants to understand how to both manage pain and recover well.
  - See how he is performing on a weekly basis.
  - Have guidance for performing these exercises alone to ensure he is performing exercises correctly.
- Fear:
  - Worried about doing something wrong which might cause pain and hinder his recovery.
  - Prescribed exercises are manageable but is unsure if he should push through discomfort.
  - \Should I push through sti ness or should I stop?"
  - \Am I bending my knee enough? Not enough?"

Name: Maya Thompson
Age: 21
Occupation: Student
Location: Hamilton, Ontario

\I was in a car accident which shattered my knee. I had to get a full knee replacement in order to continue walking. I was recently discharged from a rehabilitation facility where I started learning how to use my knee and walk again. My physiotherapist enrolled me as their patient on this application to keep up to date on my progress towards recovery."

- Goal:
  - To gain su cient use of her knee to walk and perform activities of daily living.
  - To monitor her own progress and use it as a motivating factor for pushing herself.
  - To gain enough strength in her muscles without having to further exacerbate her pain.
- Fear:
  - Her muscles need to relearn how to function and she is afraid if she performs an exercise incorrectly, the incorrect muscles will learn to compensate during the movement.
  - `Am I using the correct muscles? Is this the muscle that I need to consciously activate?'
  - Her pain might be aggravated to an intolerable degree.
  - `What if my pain does not start out as much but becomes unbearable because I did not perform the exercise correctly?'

3.6 Priorities Assigned to Users

Key users : Patients undergoing post operative physiotherapy have highest priority in the system, physiotherapists using the system to check-in and comment on patient exercises have similar priority.

3.7 User Participation

Users (Patients)

- ˆ User creates an account and set up their pro le and may sign into this account with valid credentials.
- ˆ User requests analytical information based on their past exercise inputs and results.
- ˆ User receives live form recommendations from the system by providing a video-feed of themselves performing the selected activity.
- ˆ User requests an overview of their exercise after successfully completing a video-feed upload, or looks back at old overviews.
- ˆ User creates comments on their own exercise overviews to keep personal notes and information regarding exercise session.
- ˆ User receives comments from the physiotherapist account linked to theirs.

Users (Physiotherapists)

- ˆ User creates an account and sets up their pro le, and may sign into this account with valid credentials.
- ˆ User requests a list of all patient accounts connected to their account.
- ˆ User receives overviews of patient's exercises to con rm exercise completion and progress on recovery.
- ˆ User creates comments on patient's exercise overviews to assist the patient with exercise performance, and/or to provide modi cations.

3.8 Maintenance Users and Service Technicians

Maintenance and service on the system will be performed by the current development team. Maintenance and service should be minimized with a thorough test suite to ensure all functionalities of the system are operational upon any changes or updates.

4 Mandated Constraints

4.1 Solution Constraints

Insert your content here.

4.2 Implementation Environment of the Current System

Insert your content here.

4.3 Partner or Collaborative Applications

Insert your content here.

4.4 O -the-Shelf Software

Insert your content here.

4.5 Anticipated Workplace Environment

Insert your content here.

4.6 Schedule Constraints

Insert your content here.

4.7 Budget Constraints

Insert your content here.

4.8 Enterprise Constraints

Insert your content here.

5 Naming Conventions and Terminology

5.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

Insert your content here.

6 Relevant Facts And Assumptions

6.1 Relevant Facts

Relevant facts for this project are that about a third of patients needing physiotherapy can have improved e cacy due to their exertion and form factors. This leads to slower recovery or future injury later down the line. In some instances, this can lead to a permanent decrease in an a ected joint's range of motion and strength. Additionally, there is a gap from initially being prescribed the exercise and the accountability from doing exercises on the patients own time.

The primary target of this system is for patients of physiotherapists and physiotherapists.

6.2 Business Rules

Some business rules relevant to include that there will be a distinct user interface for general users of the application and a di erent user interface for physiotherapists. A patient must be directed to use the application from the physiotherapist and will have exercises assigned to them on the system. Similarly, a physiotherapist can view and manage their patients but cannot access other patients outside their care.

A prescribed exercise chosen from the physiotherapist can have certain parameters and limitations, for example: sets, reps, frequency, etc. As well, the patient will be prompted after an exercise to give their perceived exertion as a result of the exercise from a sale of 1-10.

Furthermore, any data will not be shared without any explicit consent and stored securely.

6.3 Assumptions

Insert your content here.

7 The Scope of the Work

7.1 The Current Situation

Insert your content here.

7.2 The Context of the Work

Insert your content here.

7.3 Work Partitioning

Insert your content here.

7.4 Specifying a Business Use Case (BUC)

Insert your content here.

8 Business Data Model and Data Dictionary

8.1 Business Data Model

Insert your content here.

8.2 Data Dictionary

Insert your content here.

9 The Scope of the Product

9.1 Product Boundary

Insert your content here.

9.2 Product Use Case Table

Insert your content here.

9.3 Individual Product Use Cases (PUC's)

Insert your content here.

10 Functional Requirements

10.1 Functional Requirements

Insert your content here.

11 Look and Feel Requirements

11.1 Appearance Requirements

Insert your content here.

11.2 Style Requirements

Insert your content here.

12 Usability and Humanity Requirements

12.1 Ease of Use Requirements

Insert your content here.

12.2 Personalization and Internationalization Requirements

Insert your content here.

12.3 Learning Requirements

Insert your content here.

12.4 Understandability and Politeness Requirements

Insert your content here.

12.5 Accessibility Requirements

Insert your content here.

13 Performance Requirements

13.1 Speed and Latency Requirements

Insert your content here.

13.2 Safety-Critical Requirements

Insert your content here.

13.3 Precision or Accuracy Requirements

Insert your content here.

13.4 Robustness or Fault-Tolerance Requirements

Insert your content here.

13.5 Capacity Requirements

Insert your content here.

13.6 Scalability or Extensibility Requirements

Insert your content here.

13.7 Longevity Requirements

Insert your content here.

14 Operational and Environmental Requirements

14.1 Expected Physical Environment

Insert your content here.

14.2 Wider Environment Requirements

Insert your content here.

14.3 Requirements for Interfacing with Adjacent Systems

Insert your content here.

14.4 Productization Requirements

Insert your content here.

14.5 Release Requirements

Insert your content here.

15 Maintainability and Support Requirements

15.1 Maintenance Requirements

Insert your content here.

15.2 Supportability Requirements

Insert your content here.

15.3 Adaptability Requirements

Insert your content here.

16 Security Requirements

16.1 Access Requirements

Insert your content here.

16.2 Integrity Requirements

Insert your content here.

16.3 Privacy Requirements

Insert your content here.

16.4 Audit Requirements

Insert your content here.

16.5 Immunity Requirements

Insert your content here.

17 Cultural Requirements

17.1 Cultural Requirements

Insert your content here.

18 Compliance Requirements

18.1 Legal Requirements

Insert your content here.

18.2 Standards Compliance Requirements

Insert your content here.

19 Traceability from Requirements to Use Cases

This section maps each of the requirements to one or more of the use cases listed and de ned above in previous sections.

Table 1: Traceability from Requirements to Use Cases

Requirement	Use Case(s)
FR1	PUC3, PUC10
FR2	PUC1
FR3	PUC3
FR4	PUC10
FR5	PUC8
FR6	PUC10
FR7	PUC5
FR8	PUC7
FR9	PUC9
FR10	PUC4, PUC8
FR11	PUC11
FR12	PUC11
FR13	PUC11, PUC12
FR14	PUC6
FR15	PUC7
FR16	PUC7
FR17	PUC8, PUC10
FR18	PUC10
FR19	PUC9
FR20	PUC10
FR21	PUC13
FR22	PUC10, PUC13
FR23	PUC1, PUC13
FR24	PUC13

Continued on next page

Table 1 { continued from previous page

Requirement	Use Case(s)
FR25	PUC8, PUC10
LFR-AP1	All PUCs
LFR-ST1	PUC4, PUC6, PUC8
UHR-EOU1	All PUCs
UHR-EOU2	PUC6
UHR-LRN1	PUC5, PUC8
UHR-UPL1	PUC6
UHR-AR1	PUC5, PUC6, PUC7
PR-SL1	PUC5
PR-SL2	PUC6
PR-SCR1	PUC6, PUC7
PR-PAR1	PUC11
PR-PAR2	PUC11, PUC12
PR-RFT1	PUC5
PR-RFT2	PUC5
PR-RFT3	All PUCs
PR-CR1	All PUCs
PR-SER1	PUC11, PUC12
OER-EP1	PUC5, PUC11
OER-WE1	All PUCs
OER-WE2	All PUCs
OER-PR1	PUC1, PUC3, PUC5, PUC6, PUC8, PUC10
OER-RL1	PUC5, PUC11, PUC12
MS-MNT1	PUC13
MS-AD1	PUC5, PUC8
SR-AR1	PUC1
SR-AR2	PUC3
SR-AR3	PUC2
SR-IR1	PUC13
SR-PR1	PUC1, PUC13
SR-PR2	PUC13

2. Di erent systems may store data di erently so this is something we also have to be aware of as it could create inconsistencies when syncing the medical records.

3. When integrating the app to healthcare networks, policies and authentication layers should be carefully managed because if proper security is not in place, unauthorized users may be able to access patient data, which is extremely unsafe.

22.3 Potential User Problems

1. Learning Curve: Some users may not be the most familiar with using technology and may nd it a bit more di cult to use this app which ultimately may lead to a decrease in user satisfaction and engagement.

2. Privacy Issues and Comfortability with Video Recordings: Some users may not be comfortable with the process of recording themselves and uploading their videos to the platform, which could lead to a decrease in engagement.

3. Preference of in-person physiotherapy guidance: Some users may prefer in person appointments with their physiotherapists to receive direct feedback, which may lead to a decrease in user engagement overall.

4. Limitations of Recording Videos: Some users may nd it di cult as a result of their limited mobility to be able to record and upload the videos with precision by themselves without any additional support. This may lead to inaccurate computations and results.

5. Misinterpretation of Feedback: Users may not understand how to correctly translate or interpret the feedback given by the app, which could ultimately lead to a decrease in the e ectiveness and overall productivity of the app.

process catches integration issues early when they are less costly and disruptive to x.

15 14 Maintainability and Support Requirements

15.1 14.1 Maintenance Requirements

MS-MNT 1. The system shall automatically back up patient data once every 24 hours.

Rationale: Regular backups protect against data loss from hardware failures, software bugs, or security incidents. Daily backups balance data protection with system resource usage.

15.2 14.2 Supportability Requirements

N/A

15.3 14.3 Adaptability Requirements

MS-AD 1. The system shall support both online and limited o ine modes, caching exercise instructions for use without internet.

Rationale: Internet connectivity may be temporarily unavailable in some home environments. O ine capability with cached instructions ensures patients can continue exercises and sync data when connectivity is restored.

Fit Criterion: Exercise instructions, demonstration videos, and patient's current exercise plan are cached locally and accessible without internet connection. Cached content updates automatically when online. Users can view current exercises and instructions when o ine.

16 15 Security Requirements

16.1 15.1 Access Requirements

SR-AR 1. The system shall enforce secure login using unique credentials for each user.

Rationale: Unique credentials ensure accountability, enable proper access control, and comply with healthcare privacy regulations requiring authenticated access to patient information.

Fit Criterion: Each user account has unique credentials (username/email + password). Duplicate credentials are rejected. Authentication succeeds only with valid credentials. All access to patient data requires an authenticated session with appropriate role permissions.

SR-AR 2. The system shall enforce password rules (e.g., minimum length, complexity) and validate them before account creation.

Rationale: Strong password requirements protect against unauthorized access through brute force or dictionary attacks, safeguarding sensitive patient health information.

SR-AR 3. The system shall automatically log out users after 60 minutes of inactivity.

Rationale: Automatic logout prevents unauthorized access if a user leaves their device unattended, protecting patient privacy particularly in shared home or clinical environments.

SR-AR 4. The system shall enforce a 2FA when creating an account and logging in. The system will ensure 2FA through the email address of the user stored on the database.

Rationale: Two-factor authentication (2FA) adds an additional layer of security beyond passwords, signi cantly reducing the risk of unauthorized account access from compromised credentials. Using email-based 2FA provides a

balance between security and usability, as most users have ready access to email on separate devices. This is especially critical for protecting sensitive patient health information and ensuring compliance with healthcare data protection regulations.

SR3. The system shall include an option to reset the password if too many (5) attempts are made with incorrect login.

Rationale: Limiting login attempts to ve protects accounts from brute force attacks where attackers systematically try multiple password combinations. O ering a password reset option after failed attempts balances security with user convenience, allowing legitimate users who have forgotten their password to regain access without being permanently locked out. This threshold is high enough to accommodate normal user errors but low enough to deter automated attacks.

16.2 15.2 Integrity Requirements

SR-IR 1. The system shall encrypt all sensitive patient data at rest and in transit.

Rationale: Encryption protects patient privacy and complies with healthcare data protection regulations. It prevents unauthorized access even if data is intercepted or storage media is compromised.

16.3 15.3 Privacy Requirements

SR-PR 1. The system shall restrict patients to accessing only their own data.

Rationale: Privacy restrictions prevent patients from viewing others' medical information, complying with healthcare privacy laws and maintaining trust in the system's data protection capabilities.

SR-PR 2. The system shall ensure video and performance data storage complies with data retention and privacy requirements.

19 Compliance Requirements

19.1 Legal Requirements

19.2 Standards Compliance Requirements

20 Traceability from Requirements to Use Cases

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FR17	PUC8, PUC10
FR18	PUC10
FR19	PUC9
FR20	PUC10
FR21	PUC13
FR22	PUC10, PUC13

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